

Datathon Guide: Evaluating Model Performance vs Environmental Footprint

Lloyds Banking Group

Lloyds Banking Group is one of the largest and best-known banking organisations in the United Kingdom. It provides everyday financial services to millions of people, families, and businesses across the country, helping customers manage their money, save for the future, buy homes, and run businesses. The Group serves customers through a mix of high-street branches, online banking, and mobile apps.

The Group includes several familiar brands, such as **Lloyds Bank**, **Halifax**, **Bank of Scotland**, and **Scottish Widows**, each with its own history and customer focus. Together, these brands cover a wide range of services, including current accounts, credit cards, mortgages, savings, insurance, and pensions.

Lloyds Banking Group plays an important role in the UK economy. It supports homeowners with mortgages, lends to small and medium-sized businesses, and helps individuals plan for retirement and long-term financial security. Because of its size and reach, the Group is closely regulated and places strong emphasis on customer protection, financial stability, and responsible lending.

In recent years, Lloyds Banking Group has invested heavily in digital technology and data to improve customer experience, protect against fraud, and make banking more convenient. Its scale, trusted brands, and focus on everyday banking make it a central part of financial life for many people across the UK.

AI & ML At Lloyds Banking Group

Lloyds Banking Group uses **machine learning and artificial intelligence (AI)** to improve customer experiences, support colleagues, and make better, faster decisions across the bank. AI is embedded across its business—from detecting fraud and improving customer service to streamlining internal operations—supported by a dedicated **AI Centre of Excellence** and a strong focus on **Responsible AI** and ethical governance.

Lloyds Banking Group has been applying data science and machine learning for decades and is now expanding into **generative and agentic AI**, treating AI as a shared capability used across all business areas rather than in isolated teams. Alongside technology investment, the Group is also upskilling its workforce through training programmes and an AI Academy, ensuring AI is used safely, transparently, and at scale to help deliver its purpose of *Helping Britain Prosper*.

One major area of interest to us is **Responsible AI**; part of this is understanding the environmental impact of the hundreds of models we run every day. When building a machine learning model or AI system, we often must balance multiple factors to come to a decision about which one is best to use, including its environmental impact.

There is often a temptation to use newer, more complex models, on the assumption that they will be more performant. This has some disadvantages; complex models are often more expensive to run (both in terms of financial costs and environmental costs) and are harder to interpret.

We try to balance model performance, complexity, fairness, environmental impact, and governance requirements when selecting the right model to solve a particular problem.

In this Datathon, your mission is to choose **one of two problems**, build models, and compare predictive performance (e.g., AUC, F1, RMSE) against environmental impact (kWh and CO₂ emissions) and come to a decision about which model(s) to use and why.

Full judging criteria can be found at the end of this guide.

Good luck!

Exciting Upcoming Opportunity for Master's Students

https://lbg.wd3.myworkdayjobs.com/Undergraduate_careers/job/Bristol/Data-Science-and-AI-Masters-Placement-Scheme--Bristol-_154123



Step 1: Select Your Dataset

Choose one of the following Kaggle datasets:

[Credit Card Fraud Detection](#) (Fraud Detection)

Credit card fraud is a major financial crime issue in the UK, involving the unauthorised use of card details to make purchases or withdraw funds, often without the physical card being stolen. It commonly occurs through online “card-not-present” transactions, phishing scams, data breaches, or account takeovers, and it poses a significant challenge for both consumers and banks. UK financial institutions must detect and prevent fraud in real time while ensuring that legitimate customer transactions are not disrupted. Organisations such as **Lloyds Banking Group**, one of the UK’s largest banking groups, handle millions of card transactions every day, making fraud prevention a large-scale data and technology problem. This creates opportunities to use data analytics and machine learning to identify unusual spending patterns, adapt to new fraud techniques, and balance security with a smooth customer experience

OR

[JPX Tokyo Stock Exchange Prediction](#) (Investment Guidance)

Investment guidance in the UK focuses on helping individuals make informed financial decisions while operating within a strongly regulated environment designed to protect consumers. Rather than giving personalised advice, many modern platforms provide **guidance** – high-level information, tools, and insights that help customers understand products such as ISAs, pensions, and investments, along with the risks and trade-offs involved. UK banks and financial institutions must balance clarity, accessibility, and regulatory compliance, ensuring customers are supported without being misled or encouraged to take inappropriate risk. Organisations such as **Lloyds Banking Group** play a key role in this space, serving millions of customers with diverse financial goals, levels of experience, and risk appetites. This creates a complex challenge that blends finance, regulation, data, and user experience: how to present the right information at the right time, personalise guidance responsibly, and build trust at scale.

Download the dataset and perform basic preprocessing:

- Handle missing values
- Encode categorical variables
- Split into train/test sets

You are free to supplement your dataset with other datasets or information you can find that you believe would assist your submission.

Step 2: Choose Models from Hugging Face

Select models from the approved Hugging Face list.

Tabular Models, such as; Logistic Regression, Random Forest, XGBoost, LightGBM, CatBoost, TabNet, FB Prophet, ARIMA

Transformer Models, such as; bert-base-uncased, roberta-base, distilbert-base-uncased, microsoft/deberta-v3-base, ProsusAI/finbert, yiyanghkust/finbert-tone

T5 / Generation Models, such as: t5-small, t5-base, google/flan-t5-base, google/flan-t5-large,

LLMs, such as: TinyLlama-1.1B, google/gemma-2b, meta-llama/Llama-2-7b, mistralai/Mistral-7B

If you're not familiar with all the models, that's fine. Pick the ones you understand, and work from there.

Step 3: Train and Evaluate Performance

For classification tasks (risk/fraud), measure:

- AUC
- F1 Score
- Precision & Recall

For regression/investment forecasting:

- RMSE
- MAPE
- Directional Accuracy

For NLP tasks:

- Accuracy
- F1
- ROUGE or BLEU (for generated explanations)

Step 4: Estimate Environmental Impact

You must estimate both training and inference footprint.

Training footprint estimation formula:

$$\text{kWh} = \text{Runtime (hours)} \times \text{Average Power (kW)}$$

$$\text{CO}_2\text{e (kg)} = \text{kWh} \times \text{Grid Carbon Intensity}$$

For LLM/token-based models:

$$\text{kWh} = (\text{Total Tokens} / 1000) \times \text{kWh_per_1000_tokens}$$

$$\text{CO}_2\text{e (kg)} = \text{kWh} \times \text{Carbon Intensity}$$

Track:

- CO₂e per training run
- CO₂e per 1,000 predictions
- CO₂e per 1,000 tokens (LLMs)

Step 5: Build a Leaderboard Scoring Formula

To compare models fairly, you must design a combined performance + sustainability score.

Example formula:

$$\text{Efficiency Score} = (\text{Normalized AUC}) / (\text{Normalized CO}_2\text{e})$$

OR

$$\text{Hackathon Score} = 0.7 \times \text{Performance Score} - 0.3 \times \text{Carbon Penalty}$$

Where:

- Performance Score is normalized between 0 and 1
- Carbon Penalty increases with higher CO₂e

Your leaderboard should rank models based on this combined metric.

Step 6: Present Insights

In your final presentation, answer:

- Which model gives the best performance per kg CO₂e?
- Is a large LLM worth the carbon cost compared to smaller models?
- Does deep learning outperform gradient boosting enough to justify its footprint?
- What optimizations reduce carbon without hurting performance?

Judging Criteria

Teams will be evaluated on:

- Correct implementation (30%)
- Performance vs Carbon analysis (30%)
- Quality of leaderboard formula (20%)
- Insightfulness and recommendations (20%)